Smart Supply Chain Dashboard

KPI Reference Guide

Complete documentation of all Key Performance Indicators for Admin, Manager, and Supplier users — including formulas, thresholds, and interpretation guides.

Admin / Manager **Supplier** **AI-Powered**

Backend: FastAPI (Python) Frontend: Angular 18 Database: MongoDB Generated: July 2026

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1 Admin Dashboard KPIs

These eight KPIs are displayed on the **Admin Executive Dashboard** and are computed by the `/api/kpis/executive-summary` endpoint. They give managers a single-screen view of the entire supply chain's operational health.

1.1 Global Supply Chain Health

Display Format	Percentage (%)
API Endpoint	/api/kpis/executive-summary → metrics.health
Visible To	Admin, Manager
Good Threshold	≥ 95%

What it means: A composite indicator that summarises overall operational health by blending on-time delivery performance with inventory availability into one score.

Formula:

```
Global Health = ( OTIF% + (100 − Stockout Rate%) ) / 2
```

Example: OTIF = 94%, Stockout Rate = 2% → Health = (94 + 98) / 2 = **96.0%**

Score Range	Interpretation
≥ 95%	Excellent — operations are highly efficient
85–94%	Good — minor improvements needed
< 85%	■ Alert — immediate action required

1.2 OTIF Service Level (On-Time In-Full)

Display Format	Percentage (%)
API Endpoint	/api/kpis/executive-summary → metrics.otif
Visible To	Admin, Manager
Good Threshold	≥ 95% (world-class: ≥ 98%)

What it means: The percentage of customer sales orders that were shipped on time. An order is *on-time* when the actual shipment day does not exceed the scheduled shipment day. OTIF is the standard delivery performance metric used across global supply chains.

Formula:

```
delay = real_shipment_days − scheduled_shipment_days on-time = delay ≤ 0 OTIF% =
(on-time orders / total orders) × 100
```

1.3 Active Stockout Rate

Display Format	Percentage (%) — lower is better
API Endpoint	/api/kpis/executive-summary → metrics.stockout_rate
Visible To	Admin, Manager
Good Threshold	≤ 2%

What it means: The percentage of catalog products that currently have **zero units in stock**. A stockout means that product cannot be sold or fulfilled, directly impacting revenue and customer satisfaction.

Formula:

```
Stockout Count = COUNT(products where current_stock == 0) Stockout Rate% =
(Stockout Count / Total Products) × 100
```

1.4 Average Delivery Lead Time

Display Format	Days (d) — decimal
API Endpoint	/api/kpis/executive-summary → metrics.avg_lead
Visible To	Admin, Manager
Direction	Lower is better

What it means: The average number of days it actually takes to ship all orders. A growing average signals process bottlenecks, supplier delays, or logistics issues.

Formula:

```
Avg Lead Time = Σ real_shipment_days / Total Orders
```

1.5 Total Revenue

What it means: Total gross sales revenue generated across all orders.

Formula:

```
Order Revenue = Σ (quantity × unitPrice) per order_line Total Revenue = Σ Order Revenue across all orders (Fallback: if no line prices, estimate = order_profit / 0.15)
```

1.6 Total Orders

What it means: Total count of sales orders — a volume metric for business activity.

Formula:

```
Total Orders = COUNT(documents in sales_orders collection)
```

1.7 Active Products Catalog

What it means: Total number of distinct SKUs registered in the product catalog.

Formula:

```
Active Products = COUNT(documents in products collection)
```

1.8 Total Active Customers

What it means: Number of unique customer IDs that have placed at least one sales order.

Formula:

```
Total Customers = COUNT(DISTINCT client_id across all sales_orders)
```

2 Order & Anomaly KPIs

These KPIs are enriched by a **LightGBM machine-learning model** and rule-based logic to detect fraud and delivery anomalies. Each order from `/api/orders` is scored in real-time.

2.1 Anomaly Status (Fraud Detection)

Display Format	Badge: valid / delay anomaly / unusual
Algorithm	LightGBM classifier + rule-based threshold
API Endpoint	/api/orders (enriched per order)
Visible To	Admin, Manager

What it means: Each sales order is classified to detect suspected fraud or abnormal delivery patterns. This is the primary fraud-screening KPI powering the Sales Order tab.

Classification logic:

```
flagged "unusual" (fraud) if: LightGBM predicts fraud=1 OR: database status ==
"SUSPECTED_FRAUD" flagged "delay anomaly" if: real_shipment – scheduled_shipment >
3 days otherwise classified as: "valid"
```

LightGBM Feature Set:

Feature	Risk Direction	Description
delay_delta	↑ = Higher risk	Shipping delay beyond schedule
total_quantity	↑ = Higher risk	Unusually large order volume
total_sales	↑ = Higher risk	Unusually high revenue per order
profit_margin	↑ or ↓ = Risk	Deviation from normal margin
discount_ratio	↑ = Higher risk	Excessive discounting applied

2.2 Delay Delta

Formula:

```
Delay Delta = real_shipment_days – scheduled_shipment_days
```

Value	Status	Action
≤ 0	On Time	No action
1–3	Minor delay	Monitor
> 3	Delay Anomaly	■ Flag & investigate

2.3 Profit Margin per Order

What it means: The ratio of profit to total sales for a given order. Abnormally high or low margins are fraud signals.

Formula:

```
Total Sales = Σ (quantity × unitPrice) per order_line Profit Margin = order_profit / Total Sales
```

2.4 Discount Ratio

What it means: The average discount applied across all product lines in an order. High discount + low margin can be a fraud signal for the LightGBM model.

Formula:

```
Discounts = [product.discount for each line in order_lines] Discount Ratio = mean(Discounts) — range: 0.0 to 1.0
```

2.5 Discount–Revenue Analysis

What it means: Shows how revenue and units sold are distributed across 7 discount brackets (0%, 5%, 10%, 15%, 20%, 25%, 30%+). Reveals the relationship between promotional discounting and sales performance — critical for margin management.

Formula:

```
For each order → bin by discount_ratio into 7 brackets Revenue[bin] += total_sales Units[bin] += total_quantity
```

3 Customer Segmentation KPIs (RFM)

RFM segmentation uses **K-Means clustering (k=3)** on three standardised dimensions to classify each customer. Computed by `/api/partners/clients/segmentation`.

3.1 Recency (R)

What it means: Days since the customer last placed an order, measured from reference date **2017-12-31**. Low recency = recently active (high value). High recency = lapsed (at risk).

Formula:

```
Reference Date = 2017-12-31 Latest Order Date = max(order_date) for all orders by this customer Recency = (Reference Date − Latest Order Date).days (Default for no orders: 365 days)
```

3.2 Frequency (F)

What it means: Total number of orders placed by the customer. High frequency = loyal, repeat buyer; low or zero = one-time or lapsed customer.

Formula:

```
Frequency = COUNT(sales_orders) where client_id = this customer
```

3.3 Monetary Value (M)

What it means: Total amount the customer has spent across all orders. High monetary value customers are most important to retain.

Formula:

```
Order Value = Σ (quantity × unitPrice) per order_line Monetary = Σ Order Value for all orders by this customer
```

3.4 RFM Cluster Segments

Algorithm: Features [recency, frequency, monetary] are standardised using *StandardScaler*, then grouped into k=3 clusters using *KMeans(random_state=42)*. Clusters are labelled by sorting on average monetary value:

Rank by Avg Monetary	Label	Description
Lowest	AT RISK	Long since last purchase, low spend — churn risk
Middle	LOYAL	Regular buyers with moderate lifetime value
Highest	CHAMPIONS	High-spend, frequent, recently active customers

Bubble Chart Encoding: X = Frequency (jittered), Y = Monetary (scaled), Bubble size = Recency (larger = more recent)

4 Product & Inventory KPIs

4.1 K-Means Product Cluster Tier

API Endpoint	/api/products/clusters/summary
Algorithm	K-Means (k=3) on product features
Visible To	Admin, Manager

What it means: Products are segmented into performance tiers revealing which drive the most business value, which are steady volume drivers, and which are underperforming.

Tier	Description	Action
High Value	High revenue, premium pricing	Protect & promote
Volume Drivers	High quantity, lower margin	Optimise cost
Low Performers	Low revenue and volume	Rationalise or discount

4.2 Demand Forecast (Facebook Prophet)

API Endpoint	/api/products/forecasts?product_id=X
Model	Facebook Prophet time-series model
Visible To	Admin, Manager

What it means: Predicts future daily demand per SKU to help purchasing teams plan replenishment before stockouts occur. Auto-retrains in the background when fewer than 90 future forecast days exist.

Chart Element	Description
ACTUAL (green)	Historical daily sales
FORECAST (purple)	Predicted future demand (dashed)
Confidence Band	±5% uncertainty range around forecast

5 Supplier Dashboard KPIs

All supplier KPIs are computed by `/api/supplier/dashboard-data`. Supplier users see **only their own company's data**; Admins can select any supplier. KPIs are derived from the *purchases*, *products*, and *sales_orders* collections.

5.1 Average Lead Time

Display Format	Days (d) — decimal
Data Source	purchase_lines.supplyDelay
Visible To	Admin, Manager, Supplier
Direction	Lower is better

What it means: The average number of days between placing a purchase order and receiving goods from this specific supplier. This is the supplier's delivery speed indicator.

Formula:

```
delays = [line.supplyDelay for each purchase_line from this supplier] Avg Lead Time = sum(delays) / count(delays) (Default: 10.0 days if no data)
```

5.2 OTIF Rate (Supplier-Level, 12-Day Threshold)

Display Format	Percentage (%)
On-Time Threshold	supplyDelay ≤ 12 days
Visible To	Admin, Manager, Supplier
Good Threshold	≥ 90%

What it means: The percentage of individual purchase order lines delivered within the 12-day threshold. Unlike the Admin OTIF (which measures customer shipment), this measures the **supplier's own reliability** in delivering goods to the warehouse.

■ **Note:** The Admin OTIF uses *scheduled vs. real shipment* to measure customer delivery. The Supplier OTIF uses *supplyDelay ≤ 12 days* to measure inbound procurement performance. These are two distinct KPIs.

Formula:

```
on_time_count = COUNT(purchase_lines where supplyDelay ≤ 12) total_lines = COUNT(all purchase_lines from this supplier) Supplier OTIF% = (on_time_count / total_lines) × 100
```

5.3 Total Volume Supplied

What it means: Total number of units delivered by this supplier across all purchase orders. Primary throughput metric for supplier performance evaluation.

Formula:

```
Total Volume = Σ line.quantity for all purchase_lines from this supplier
```

5.4 Total Supply Cost

What it means: Total monetary value of all goods purchased from this supplier (procurement cost, not sales revenue). Used to evaluate procurement spend concentration.

Formula:

```
Total Supply Cost = Σ (line.quantity × line.unitPrice) for all purchase_lines
```

5.5 Total Sales Revenue (Downstream)

What it means: Revenue from customer sales orders containing products supplied by this supplier. Links upstream supply to downstream demand, showing business dependency.

Formula:

```
supplier_skus = SET of product SKUs in this supplier's purchase lines Downstream
Rev. = Σ (line.quantity × line.unitPrice) for matching SKUs in sales_orders
```

5.6 Total Sales Volume (Downstream)

What it means: Total units sold downstream to customers for products this supplier provides. Helps the supplier understand demand pressure on their products.

Formula:

```
Downstream Volume = Σ line.quantity for all matching order_lines (supplier SKUs
only)
```

5.7 Active Sales Orders Count

What it means: Count of customer sales orders containing at least one product from this supplier. Represents the backlog that depends on this supplier's stock availability.

Formula:

```
Sales Orders Count = COUNT(sales_orders where any order_line.product_sku ∈
supplier_skus)
```

5.8 Low Stock Count

What it means: Number of supplier products whose current stock has fallen **below the Reorder Point**. Products in this state are at risk of stockout and require urgent replenishment.

Formula:

```
Low Stock Count = COUNT(products where current_stock < reorder_point)
```

5.9 Safety Stock

Display Format	Integer (units)
Formula Type	Statistical — Z-score method
Service Level	95% (Z = 1.65)
Minimum Floor	15 units
Visible To	Admin, Manager, Supplier

What it means: A buffer stock level held to protect against unexpected demand spikes or supply delays. It is the minimum inventory that must always be maintained to avoid stockouts during the replenishment lead time.

Formula:

```
Safety Stock = Z × σ_demand × √(Lead Time) Safety Stock = max(15, Safety Stock) ← minimum floor Where: Z = 1.65 (95th percentile of normal distribution) σ_demand = std dev of historical daily demand for this SKU Lead Time = average supply delay in days (from supplyDelay)
```

5.10 Reorder Point (ROP)

Display Format	Integer (units)
Chart	Orange dashed line on Stock Level bar chart
Trigger	current_stock < ROP → product flagged as Low Stock
Visible To	Admin, Manager, Supplier

What it means: The stock level at which a new purchase order *must* be placed to avoid a stockout before the next delivery arrives. When stock drops below this level, the product is added to the replenishment alert list.

Formula:

```
ROP = Safety Stock + (Mean Daily Demand × Lead Time)
```

5.11 Mean Demand & 5.12 Standard Deviation of Demand

Mean Demand is the average daily sales quantity for a given SKU, derived from historical sales and Prophet forecast records. It is the core demand signal for all stock calculations.

Formula:

```
demand_samples = [f.sales or f.forecast for each forecast record where product_id = sku] Mean Demand = sum(demand_samples) / count(demand_samples) σ_demand = √[ Σ(demand_i − mean_demand)² / n ] (population std dev)
```

5.13 Suggested Reorder Quantity

What it means: The recommended quantity to order in the next replenishment to bring stock back to a target level covering 2 full lead-time cycles plus safety stock. Ensures both the current deficit is covered and a future buffer is established.

Formula:

```
Target Stock = Safety Stock + (2 × Mean Demand × Lead Time) Suggested Reorder = max(0, Target Stock − Current Stock)
```

6 Supplier Segmentation KPIs

Displayed on the **Segmentation** tab. Computed by `/api/partners/suppliers/segmentation` using K-Means (k=3) on [avg_delay, total_volume, total_cost]. Clusters are ranked by *average total volume*.

Cluster Label	Basis	Description
UNDERPERFORMING	Lowest volume	Small-scale or unreliable supplier; high delays
RELIABLE	Middle volume	Consistent mid-tier supplier with acceptable delays
STRATEGIC	Highest volume	High-volume, critical partner for the supply chain

KPI	Formula	Direction
Avg Delay	mean(supplyDelay) per supplier	Lower is better
Total Volume	Σ line.quantity per supplier	Higher is better
Total Cost	Σ (line.qty × line.unitPrice) per supplier	Track for spend control

7 AI Narrative KPIs (LLM Executive Summaries)

Both dashboards use a local **Ollama LLM** (preferred model: `qwen2.5:7b`, fallbacks: `llama3.1`, `mistral`) to generate natural-language summaries from numeric KPI data.

Dashboard	Inputs to LLM	Output
Admin	Global Health%, OTIF%, Stockout Rate%, Avg Lead Time, Revenue, Orders, Products, Customers	3-sentence analysis of operational health, stock & delivery, and recommended focus areas
Supplier	Avg Lead Time, Supplier OTIF%, Volume Supplied, Low-Stock Count, Downstream Orders Count	3–4 sentence letter to the supplier on delays, downstream impact, and restock urgency

If the LLM is unavailable, the system falls back to a dynamically-generated template summary that populates the same narrative fields with live metric values.

8 Quick Reference Card

Admin KPIs at a Glance

KPI	Formula	Good Threshold
Global Health	(OTIF% + (100 – Stockout%)) / 2	≥ 95%
OTIF Service Level	on-time orders / total orders × 100	≥ 95%
Active Stockout Rate	products (stock=0) / total products × 100	≤ 2%
Avg Delivery Lead Time	Σ real_shipment / total orders	As low as possible
Total Revenue	Σ (qty × unitPrice) per order_line	Trending up ↑
Total Orders	COUNT(sales_orders)	Trending up ↑
Active Products	COUNT(products)	Stable
Total Customers	COUNT(DISTINCT client_id)	Trending up ↑

Order Anomaly KPIs at a Glance

KPI	Formula / Logic	Threshold
Anomaly Status	LightGBM predict(features) + status flag	unusual = fraud risk
Delay Delta	real_shipment – scheduled_shipment	> 3d = anomaly flag
Profit Margin	order_profit / total_sales	Stable = normal
Discount Ratio	mean(product.discount) per order_line	Outlier = risk signal

Supplier KPIs at a Glance

KPI	Formula	Good Threshold
Avg Lead Time	mean(supplyDelay) across purchase_lines	As low as possible
Supplier OTIF Rate	lines (delay ≤ 12d) / total_lines × 100	≥ 90%
Safety Stock	1.65 × σ_demand × √(Lead Time) [floor=15]	Always above 0
Reorder Point (ROP)	Safety Stock + (Mean Demand × Lead Time)	stock > ROP always
Suggested Reorder	max(0, TargetStock – CurrentStock)	Order when flagged

RFM Segmentation at a Glance

Segment	Recency	Frequency	Monetary	Retention Action
CHAMPIONS	Low (recent)	High	High	Reward & upsell
LOYAL	Medium	Medium	Medium	Engage & grow
AT RISK	High (lapsed)	Low	Low	■ Reactivation campaign

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